

Novel operational strategies to maximize crop and electricity production in single-axis agrivoltaic systems based on light response curve and daily light integral

Abstract

Dynamic Agrivoltaic (APV) systems require tracking strategies that reconcile crop light demand with electricity generation, yet conventional True Tracking (TT) maximizes energy at the expense of ground irradiance. This study proposes two novel control strategies for single-axis APV systems: Daily Light Integral Tracking (DLIT) and Knee-Point Tracking (KPT). These strategies embed crop physiological requirements into tilt control using stage-dependent Daily Light Integral (DLI) profiles and temperature-sensitive Light Response Curves (LRCs). A simulation framework combining solar geometry, ground irradiance modeling, Leaf Area Index (LAI) dynamics, and photosynthetic carbon assimilation was applied to barley under Swedish conditions and benchmarked against TT, anti-tracking (AT), and scheduled tracking (ST). Relative to TT, KPT increased biomass from 10.60 to 11.67 ton ha⁻¹ (+10.1 %) while reducing energy yield from 937.74 to 804.72 kWh kWp⁻¹ (-14.2 %). DLIT achieved near-optimal electricity production (933.94 kWh kWp⁻¹, -0.4 % vs. TT) with a modest biomass gain to 10.76 ton ha⁻¹ (+1.5 %), demonstrating its ability to prioritize energy conversion after meeting crop DLI targets. Smoothing of tilt trajectories reduced mechanical oscillations but affected strategies differently: biomass decreased by 22–24 % across cases, while energy impacts varied (-8.5 % TT, +19.6 % AT, -7.5 % DLIT, -0.6 % KPT) due to distinct baseline tilt-angle dynamics. The results show that crop-informed tracking enhances land-use efficiency by dynamically balancing food and energy objectives. Although quantitative outcomes are crop- and site-specific, the DLIT and KPT frameworks are transferable to other crops when appropriate LAI development, DLI targets, and light-response parameters are provided, enabling scalable next-generation APV control.

1. Introduction

The increasing global demand for both energy (47 % by 2050) and food (35-56 % by 2050) represents a critical challenge for sustainable development [1,2]. Solar Photovoltaic (PV) technology has experienced unprecedented growth in recent years and is expected to lead renewable capacity expansion through 2030, with solar PV alone projected to account for approximately 80 % of new renewable power capacity additions worldwide according to the International Energy Agency's latest analysis, highlighting the pivotal role of solar in the clean energy transition [3]. Agrivoltaic (APV) systems, which co-locate crop production with solar PV conversion on the same land, have been identified as a promising approach to address competing land-use demands by enhancing land-use efficiency and contributing to both food and energy security [4], [5], [6], [7], [8], [9]. Hence, APV systems contribute to key United Nations Sustainable Development Goals, including food security (SDG 2), affordable and clean energy (SDG 7), and climate action (SDG 13) [10], [11], [12], [13], [14]. APV has progressed beyond experimental trials, with more than 14 GW_p of installed capacity worldwide, demonstrating its emergence as a mainstream component of renewable energy infrastructure [15,16].

PV systems equipped with solar trackers are significantly more efficient than fixed-tilted systems, achieving energy yields up to 24 % higher [17,18]. This increase in performance contributes to a lower Levelized Cost of Electricity (LCOE), making tracking systems more economically attractive [19,20]. Garrod and Ghosh [21] conducted a comparative study on fixed-tilted, single-axis, and dual-axis tracking configurations across diverse regions, including Asia, Europe, North America, and Australia, and found that single-axis tracking consistently delivered the lowest LCOE. Another study estimated that single-axis APV systems across Europe achieved a 20–25 % lower LCOE compared to vertical APV systems, and approximately 30 % lower than overhead APV configurations [19]. Tracking systems improve energy conversion by dynamically adjusting the orientation of PV modules to follow the sun's path, thereby minimizing the angle of incidence and maximizing solar capture [22,23]. However, in APV systems, this efficiency gain comes with trade-offs. Increased ground shading from dynamic tracking structures alters the Photosynthetically Active Radiation (PAR) direct and diffuse components to crops [24]. This can adversely affect crop growth, depending on factors such as crop type, local weather, and site-specific growing conditions. Therefore, the integration of solar tracking in APV systems requires careful consideration of both energy and agricultural productivity to ensure an optimal balance between the two.

In APV systems utilizing single-axis solar trackers, it is possible to modulate irradiance levels both at the crop level and on the PV modules by alternating the tracker's

operational strategy [25]. This can be achieved by switching between True Tracking (TT), which follows the sun's path throughout the day (excluding early morning and late afternoon to prevent inter-row shading between PV arrays), and Anti-Tracking (AT), which intentionally deviates from the sun's trajectory to reduce direct sunlight exposure. This dynamic adjustment allows for optimized light distribution, balancing energy conversion with crop growth requirements in a controlled and efficient manner [25], [26], [27], [28], [29], [30], [31], [32], [33].

Previous studies have developed operational strategies for single-axis solar trackers in APV systems and compared the difference in energy and crop yield between the TT and AT strategies. The most common solar tracking strategy in APV is to combine TT and AT based on the time of day, using rule-based approaches. Valle et al. [26] implemented TT between 11:00 and 15:00, using AT for the remaining hours of the day. Similarly, Elamri et al. [29] applied TT from 9:00 to 14:00, citing the need to reduce heat stress at the crop level. Alam and Butt [33], Willockx et al. [28], and Riaz et al. [30] implemented controlled tracking strategies that apply AT during certain hours before and after midday, prioritizing the crop growth, and TT around midday to maximize energy output. Knapp and Sturchio [27] also divided each day into an AT period before midday and a TT period afterward, explaining that plants exposed to morning sunlight tend to produce more than those that primarily receive afternoon light. Grubbs et al. [25] proposed a controlled tracking strategy applied during specific day intervals in the year, corresponding to critical crop growth stages. These intervals were grouped into three clusters: early vegetative growth, late vegetative growth, and early to late reproductive periods. Wang and Sun [31] introduced a novel controlled tracking approach based on the crop's light saturation point, a key physiological threshold indicating the level of light intensity beyond which photosynthesis no longer increases. Their method uses Photosynthetic Photon Flux Density (PPFD) as a real-time indicator of light availability to determine the tracking strategy. AT is applied when PPFD values range from 50 % to 100 % of the saturation point, ensuring crops receive sufficient light without excess exposure. When PPFD falls below 50 % or exceeds 100 %, the system switches to TT to either enhance light interception or increase ground shading.

Bruno et al. [32] proposed an optimization-based tracking strategy that differs from previous approaches by directly integrating monthly, crop-specific light requirements into the tracking algorithm. Their method builds on field research from the Landwirtschaftliches Technologiezentrum (LTZ) in apple orchards, which established acceptable daily light reduction thresholds for each month. These thresholds serve as irradiance targets for the apple trees. The system dynamically adjusts the tilt angle of the solar modules to balance energy conversion with crop light availability. By simulating a range of tilt configurations, the model aims to closely align the actual irradiance reaching the tree canopy with the LTZ-defined targets. This approach

represents a shift from generalized tracking strategies toward more precise and data-driven systems informed by long-term agronomic insights.

[Table 1](#) summarizes the current state of the art in APV tracking strategies reported in the literature. The table consolidates key characteristics of existing approaches in a concise and structured format, enabling direct comparison of control logic, crop awareness, and reported limitations across studies.

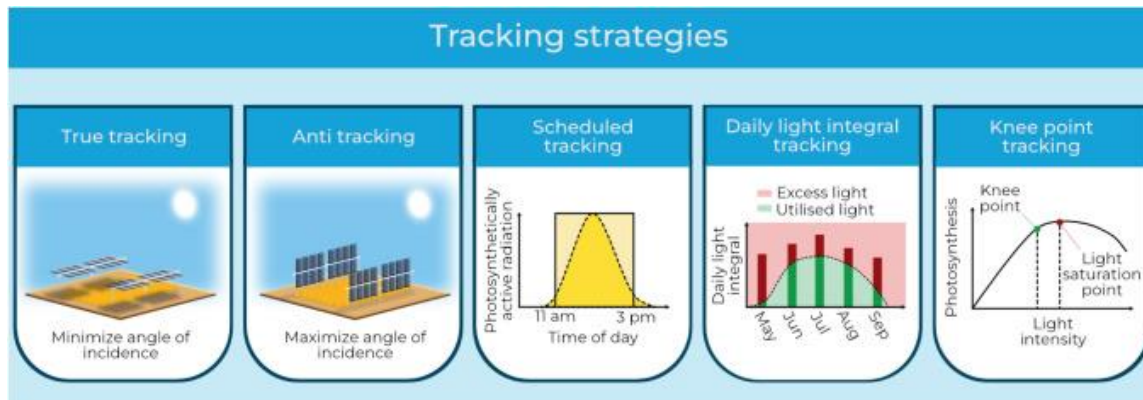
Table 1. Summary of previous study on the tracking system for APV.

Reference	Tracking Strategy	Control Logic	Crop Aware Tracking	Key Contribution	Limitation
Valle et al. [26]	Time-based TT + AT	TT (11:00–15:00), AT otherwise	No	Introduced time-segmented TT/AT tracking	No physiological crop input
Elamri et al. [29]	Time-based TT + AT	TT (09:00–14:00), AT otherwise	No	Introduced time-segmented TT/AT tracking	Fixed hourly thresholds
Alam & Butt [33]	Controlled AT before/after midday	Crop-priority AT morning/evening, midday TT	No	Introduced time-segmented TT/AT tracking	No dynamic light feedback
Willockx et al. [28]	Scheduled TT/AT	Time-window control	No	Introduced time-segmented TT/AT tracking	Static scheduling
Riaz et al. [30]	Controlled TT/AT	Crop-priority morning/evening shading	No	Balanced crop protection and energy yield	No physiological crop modeling
Knapp & Sturchio [27]	Morning AT, afternoon TT	Morning-light crop AT, afternoon TT	No	Introduced time-segmented TT/AT tracking	Not crop-specific
Grubbs et al. [25]	Seasonal stage-based tracking	TT/ AT aligned with growth stages	No	Linked tracking strategy to crop phenology/seasonal tracking	No real-time feedback
Wang & Sun [31]	PPFD-based adaptive tracking	Switch TT/AT based on light saturation	Yes	Introduced crop light-saturation-driven control	Limited to saturation threshold logic

Bruno et al. [32]	Optimization-based crop-target tracking	Monthly irradiance targets	Yes	Integrated agronomic irradiance thresholds into tracking	No Physiological crop modeling
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Previous APV tracking studies, including those by Bruno et al. [32] and Wang and Sun [31], have not evaluated the direct impact of tracking strategies on crop yield and instead rely on pre-determined, empirically derived thresholds of acceptable PAR reduction obtained from external experimental facilities. As these thresholds are not generated within a dynamic modeling framework, such approaches remain less generalizable and lack the ability to estimate crop-specific tolerance to PAR reduction under variable conditions. More broadly, existing APV tracking strategies predominantly rely on fixed schedules, heuristic TT/AT switching, or empirically defined irradiance limits, while crop physiological light demand is rarely incorporated as an explicit control variable in tracker tilt algorithms.

To address this methodological limitation, the present study introduces a crop-informed tracking framework based on mechanistic representations of crop light response. Two novel operational strategies for single-axis trackers are explored. The first strategy uses the crop-specific Daily Light Integral (DLI) as a decision variable to determine when to apply TT or AT, while the second identifies the Knee Point (KP) of the crop's Light Response Curve (LRC), representing the onset of photosynthetic saturation, to guide optimal tilt-angle selection. Although previous studies have reasonably assumed that midday irradiance often exceeds crop light demand, this assumption does not fully account for weather-induced irradiance variability, such as cloud cover, nor do most studies consider the risk of photoinhibition under excessive irradiation. These limitations highlight the absence of a formal control mechanism capable of dynamically regulating tracker operation based on real-time crop light response. Accordingly, the proposed Daily Light Integral Tracking (DLIT) and Knee-Point Tracking (KPT) strategies enable dynamic prioritization between energy conversion and crop growth by ensuring that crops receive sufficient, but not excessive light. A Scheduled Tracking (ST) strategy with fixed TT/AT time windows is also included as a benchmark, reflecting commonly adopted approaches in prior APV studies and a schematic illustration of all strategies is provided in Fig. 1.



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Fig. 1. Schematic representation of the tracking strategies evaluated in this study. TT, AT, and ST are included as benchmarks against conventional methods and previous studies, while DLIT and KPT represent the novel strategies introduced in this work.

The remainder of this paper is structured as follows: [Section 2](#) outlines the methodology, detailing the tracking strategies and evaluation criteria; [Section 3](#) presents the results, highlighting key findings and comparative analyses; [Section 4](#) provides an in-depth discussion of the implications and limitations of the study; finally, [Section 5](#) concludes the paper by summarizing the main contributions and proposing directions for future research.

2. Methodology

This section outlines the site characteristics and PV system configuration used in the study, details the modeling framework that supports the tracking strategies, and introduces the novel operational approaches proposed.

This study employs a Python-based model, Agri-OptiCE®, to simulate and evaluate the energy and biomass outcomes of five solar tracker operational strategies for APV systems for the crop barley. The five strategies are TT, AT, ST, DLIT, and KPT. The TT, AT, and ST are methods that have been used in previous studies and are used for benchmarking to the two newly developed approaches in this study, the DLIT and KPT. A short description of the operational strategies, including the benchmarks are as follows:

TT – The standard operation of the solar tracker, designed to maximize energy output.

AT – The inverse of TT, aimed at maximizing ground irradiance or minimizing solar irradiation interception by the PV system.

ST – A hybrid strategy that alternates between TT and AT throughout the day. In this approach, TT is applied from 11:00. to 15:00, while AT is used during the remaining hours. This schedule is based on Valle et al. [26], who identified this midday period as both the hottest and the most irradiated, conditions under which incoming light may exceed the crop’s usable range.

DLIT – Initially follows the AT strategy until the crop's required DLI is met, then switches to TT for the remainder of the day prioritizing energy conversion.

KPT – It chooses the optimal tilt angle based on the ambient temperature in the field that is derived out of the optimal PAR of the LRC using the KP.

The performance of DLIT and KPT is analyzed and compared to the established outcomes of TT, AT, and ST to assess their relative effectiveness in optimizing both energy capture and biomass production. Table 2 presents the APV system component and specification used in this study concisely.

Table 2. Primary APV system components and specifications.

Component	Specification	Notes
Location	Kärro Prästgård, Västerås, Sweden (59.55° N, 16.76° E)	Experimental APV research site
PV configuration	Single-axis tracking APV system	Tracker axis aligned with field layout
Installed capacity	26.1 kW _p	Simulated system
PV module type	Monocrystalline bifacial silicon	JAM72-D-40_580-LB
Module rated power	580 W _p	STC
Bifacial factor	0.85	Manufacturer specification
Number of modules	45	Three tracker rows
Row length	20 m	Per tracker row
Row spacing	10 m	Row-to-row distance

Tracker hub height	1.8 m	Clearance for agricultural operations
Tracker rotation range	$\pm 90^\circ$	Matches on-site tracker limits
Inverter (simulation)	30 KW	SUN2000-30KTL-M3-380
Meteorological inputs	GHI, DHI, PAR, temperature, wind speed, pressure	Hourly data (2019)
Crop reference	Barley (<i>Hordeum vulgare</i> L.)	Representative cereal crop

2.1. Site characteristics and PV system

The choice of a single-axis architecture is motivated by the existing APV infrastructure at the Kärrobo Prästgård research site near Västerås, Sweden (59.55°N , 16.76°E), which hosts the country's first operational APV facility. The site includes several APV system configurations running in parallel, notably a single-axis tracking APV system with an installed capacity of 26.1 kW_p . This system features three rows of PV modules, each 20 meters long and spaced 10 meters apart. The system is mounted at a hub height of 1.8 meters. A total of 45 silicon monocrystalline bifacial PV modules (model JAM72-D-40_580-LB) with a bifacial factor of 0.85, each rated at 580 W_p , are used in the setup. To enhance irradiation transmittance for optimal crop growth, the solar tracker's rotation angle is constrained to $\pm 90^\circ$, aligning with the current practical tracking configuration at the Kärrobo Prästgård site. This alignment ensures consistency between the simulated tracking framework and the on-site system, supporting the realism and future experimental verification of the proposed DLIT and KPT strategies. Hourly weather data for 2019 is collected from the Swedish Meteorological and Hydrological Institute (SMHI), including Global Horizontal Irradiance (GHI), Diffuse Horizontal Irradiance (DHI), ambient temperature, wind speed, atmospheric pressure, and PAR [34].

2.2. Model design

The model employed in this study builds upon the Agri-OptiCE® framework, originally developed in MATLAB® as the open-source package Opti-CE [35], Campana et al. [36], and Campana et al. [37] and has been further implemented and extended in Python by Ma Lu et al. [38], and Zainali et al. [39]. The functions used from the previous model, Agri-OptiCE®, are mainly the solar position, direct and diffuse shading, and hourly power generation [[37], [38], [39]]. Based on direct and diffuse shading calculations, the PV tilt angle is optimized at each hourly step using the Tilt Angle Scanning Method, a discrete tilt optimization technique, to either maximize ground irradiance or maximize

the irradiation intercepted by the PV system. The objective function at each time step , finds the tilt angle that maximizes or minimizes the ground irradiance as shown in Eq. (1). The ground irradiance is calculated in the Agri-OptiCE® as follows:(1)where is the global horizontal irradiance (W m^{-2}), is the diffuse horizontal irradiance (W m^{-2}), is the view factor for direct irradiance, and is the view factor for diffuse irradiance [40].

2.3. Novel operational strategies

2.3.1. Daily light integral tracking (DLIT)

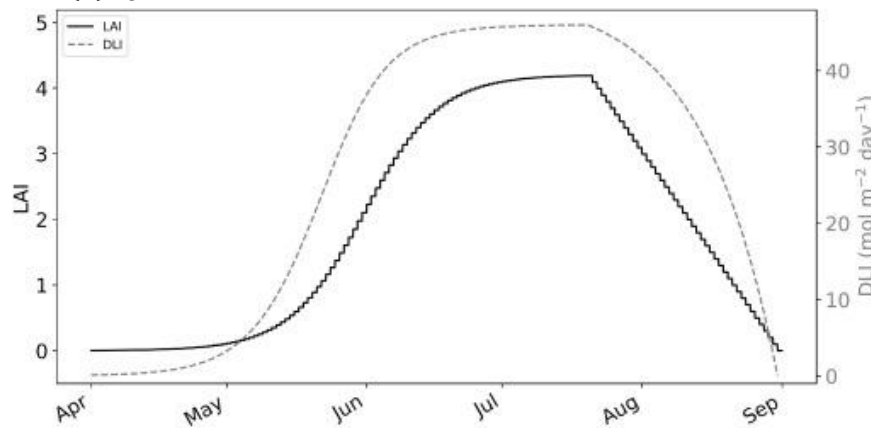
This solar tracker operational strategy prioritizes ground irradiance until the DLI requirement of a given crop is satisfied for a specific day, after which PV energy interception is optimized. The strategy implements AT until the DLI of a specific crop is met and TT for the rest of the day. Since the meteorological data available for the assumed site Kärrobo Prästgård is hourly, the conversion of the ground irradiance in GHI (W m^{-2}) to DLI ($\text{mol m}^{-2} \text{ day}^{-1}$) is given by [41,42]:(2)where (hourly), and the second term can be calculated from GHI as follows:(3)

is the GHI to PAR conversion factor, assumed to be 0.44 [41] and the PAR in W m^{-2} is converted to a $\mu\text{mol m}^{-2} \text{ s}^{-1}$ using a factor 4.57 [42]. Accordingly, Eq. (3), which quantifies the cumulative PAR derived from the GHI, can be substituted into Eq. (2) to compute the corresponding DLI. However, in this case, the accumulated DLI up until an hour of the day is calculated in Eq. (4). Thus, is defined as:(4)

To calculate the DLI requirement profile of barley throughout its growth cycle, the Leaf Area Index (LAI) is modeled using a sigmoid function, calibrated to the maximum LAI typically observed in barley crops in northern Europe under open field condition. A logistic growth function captures the biological development of the plant, starting with slow leaf emergence in the early stages, followed by rapid leaf expansion during mid-growth, and tapering off as the plant approaches maturity. After reaching its peak, the leaf area gradually declines through a linear reduction, representing natural senescence leading up to harvest. The sigmoid function used to calculate the S-shaped LAI profile is as follows [43,44]:(5)where is the number of days after planting, is the maximum LAI during the plant growth period, is the mid-way to the maximum growth stage of LAI, and is the logistic growth rate of 0.12 day^{-1} when adjusted for the effective photoperiod excluding non-photosynthetic nighttime hours [45].

The used in this study, Eq. (5), is based on the findings of Nowak et al. [46], who conducted experiments on two barley varieties across two growing seasons with varying sowing densities and contrasting rainfall conditions, one dry and one wet. For this

analysis, the κ is set at 4.2 and the LAI profile for the whole barley growth period is presented in Fig. 2. Then based on the LAI profile, the DLI profile is then calculated using Beer-Lambert law that serves as a DLI target for every day. By determining the fraction of absorbed light by the barley canopy for every day, the DLI is calculated as follows [47,48]: (6) where $DLI(x)$ is the DLI for any given day denoted by x days after planting, DLI_{max} is the maximum DLI for barley, κ is the extinction coefficient and LAI is the leaf area index for the number of days after planting as indicated in Eq 6. The DLI_{target} can be defined on a daily basis or assumed as a monthly average. A previous study by Bruno et al. [32] used a monthly tolerated reduction in irradiation; similarly, the DLIT strategy proposed here can be tested using either a daily variable $DLI(x)_{target}$ or a monthly averaged $DLI(x)_{target}$.



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Fig. 2. LAI and DLI profile of the crop barley for the whole growth period.

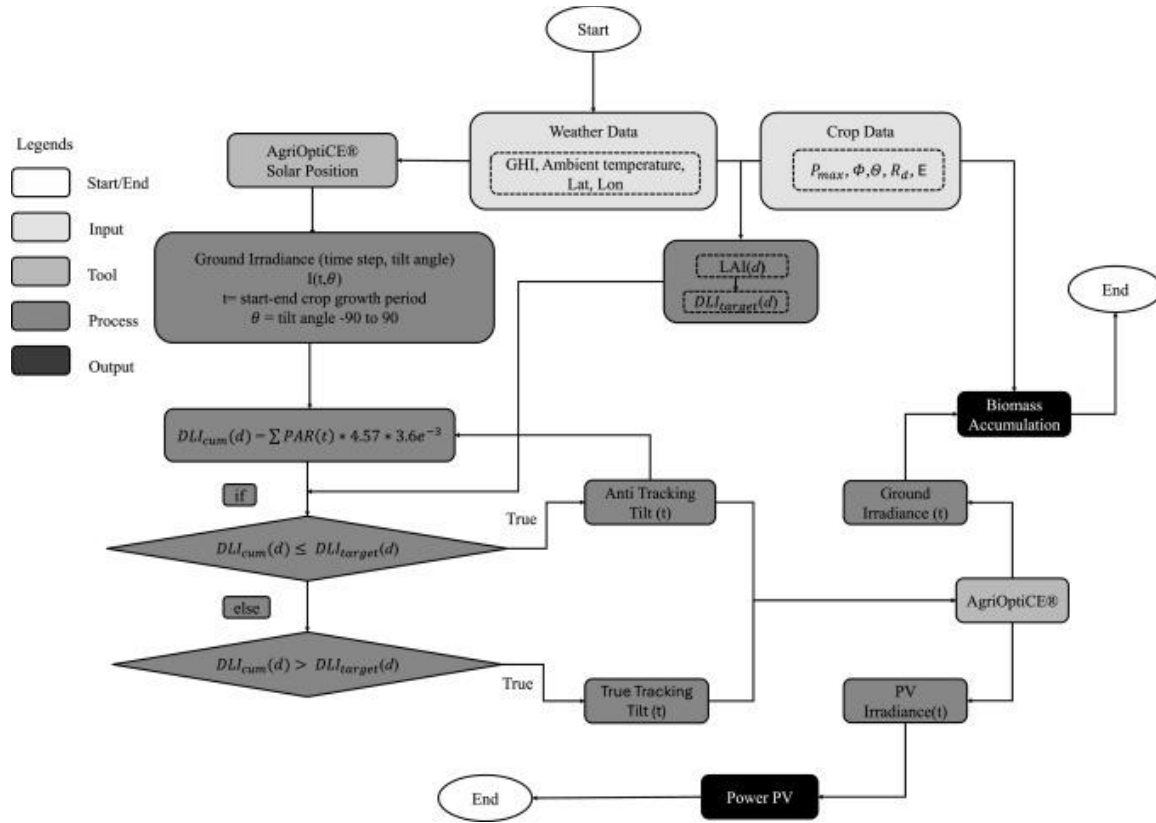
A reasonable estimate for the maximum DLI_{max} was obtained from the literature by Burnett et al. [49], with a value of $25.9 \text{ mol m}^{-2} \text{ day}^{-1}$. This value represents the upper limit of cumulative PAR typically intercepted by a well-developed barley canopy under optimal environmental and physiological conditions. It serves as a reference point for scaling the simulated DLI values throughout the crop growth cycle.

Therefore, the operational strategy continuously monitors the hourly accumulation of DLI and dynamically shifts between AT and TT once the accumulated DLI meets or exceeds the daily target. The tracking strategy is determined as AT, if $DLI(x) \leq DLI_{target}$ and TT, if $DLI(x) > DLI_{target}$.

The DLI profile shown in Fig. 2 represents DLI_{target} that serves as a decision threshold to determine whether the system should prioritize crop growth or energy conversion. On each day, the system gives priority to the crop until the DLI_{target} for that specific day is achieved. Once the target is met, the system then shifts its focus to maximizing energy conversion. The maximum DLI target of $25.9 \text{ mol m}^{-2} \text{ day}^{-1}$, adopted from Burnett et

al. [49], is assumed to correspond to the reproductive stage of C3 cereals when LAI and photosynthetic demand are highest; the plateau observed in Fig. 2 reflects this stage of maximum light requirement, while lower targets are applied during earlier vegetative development.

This approach ensures that the light needs of the crops have high priority. Fig. 3 illustrates the schematic flowchart outlining the simulation process used to implement the DLIT method, providing a clear overview of the key computational steps involved.



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Fig. 3. Schematic diagram showing the flow of the DLIT model.

2.3.2. Knee-point tracking (KPT)

The KP detection method can be used to determine the optimal level of PAR on the LRC, as previously suggested by Ma Lu et al. [50], helping to determine the ideal or closest possible tilt angle that aligns with this optimal PAR. At each time step the incoming PAR is compared to the optimal PAR and the tilt angle that yields the closest to the optimal PAR is selected as optimal tilt. This optimal PAR corresponds to the point on the LRC

where the photosynthetic rate begins to plateau. To identify this KP, a straight line is drawn between the start and end points of the LRC. Perpendicular lines are then drawn from each point on the curve to this baseline. The point with the longest perpendicular distance represents the "KP," the optimal point on the curve for selecting the tilt angle [51,52].

LRC describes the relationship between photosynthetic rate and PAR, where the photosynthetic rate is calculated for every PAR in a range from 0 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ and every temperature in the range 0 to 35°C using the non-rectangular hyperbola equation, presented in Eq. (7), which accurately models the response of photosynthesis rate to light intensity [[53], [54], [55], [56], [57], [58]]:

$$P = \frac{P_{\text{max}}}{1 + \frac{P_{\text{max}}}{I_0} + \frac{R_d}{I_0} + \frac{R_d}{I_0} \exp\left(-\frac{I_0}{P_{\text{max}}}\right)}$$

where P is the maximum rate of fixation ($\mu\text{mol m}^{-2} \text{s}^{-1}$), I_0 is the maximum quantum yield of (mol CO₂ mol⁻¹photons), determines the curvature of the function, R_d is the dark respiration ($\mu\text{mol m}^{-2} \text{s}^{-1}$) and I is the PAR ($\mu\text{mol m}^{-2} \text{s}^{-1}$).

However, the P_{max} , I_0 , and R_d are temperature dependent. The hyperbola equation can be directly used if only the temperature is fixed. The simulated environment, an agricultural field, does not presume constant temperature, and these parameters must be calculated as temperature dependent. Arrhenius function normalized with respect to 25°C can be used to calculate the parameters that are placed in hyperbola equation, corresponding to the ambient temperature of the simulated site [59].

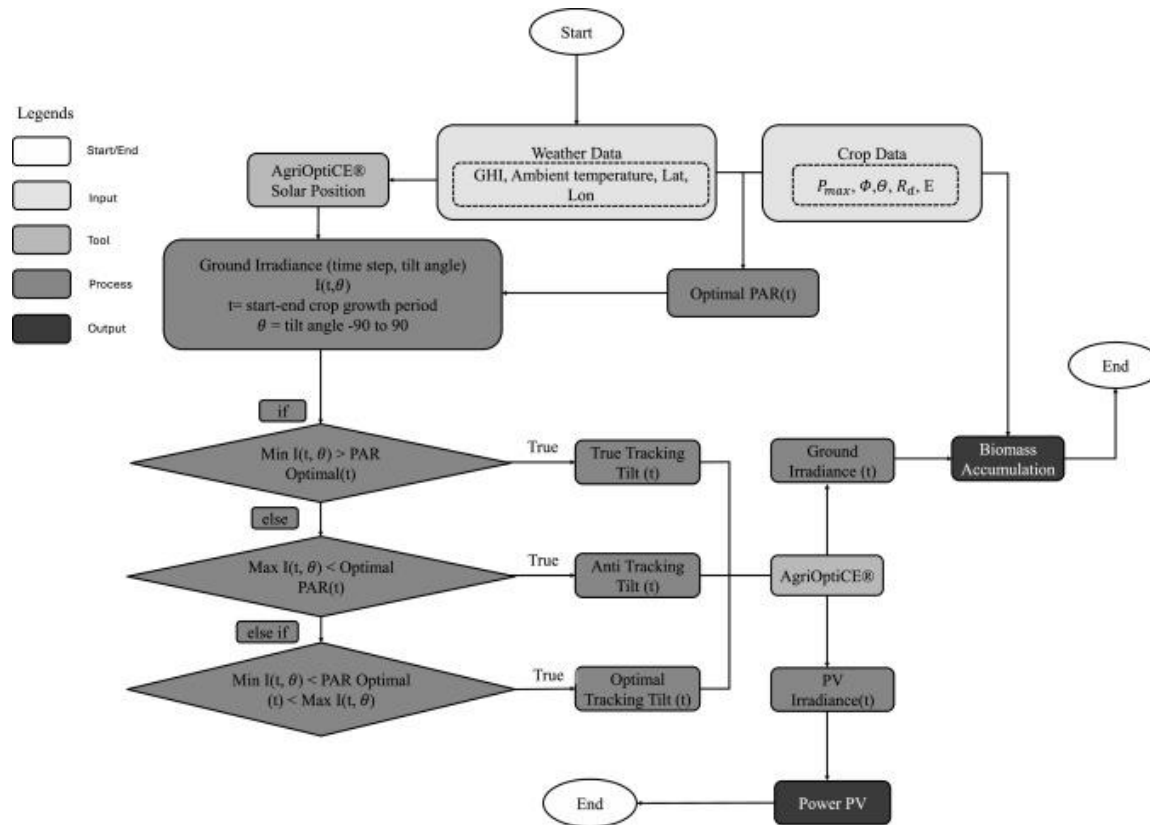
$$P = \frac{P_{\text{max}}}{1 + \frac{P_{\text{max}}}{I_0} + \frac{R_d}{I_0} + \frac{R_d}{I_0} \exp\left(-\frac{I_0}{P_{\text{max}}}\right)}$$

where P is any of the parameters P_{max} , I_0 , and R_d , T is the temperature at every hour step in °C, E_a is the activating energy (J mol⁻¹), and R is the universal gas constant (8.314 J mol⁻¹ K⁻¹).

The parameter values for P_{max} , I_0 , and R_d at 25°C can be obtained from studies [53,60]. Their temperature dependence is determined using Eq. (8), and the resulting temperature-adjusted values are then applied in Eq. (7) to compute the net photosynthetic rate. Activation energy 'E' is the minimum energy required to start a biochemical reaction of carbon fixation, a key factor for the temperature sensitivity of photosynthetic processes [61]. In this study, an interpolation of the values provided by Nolan and Smillie [62] was used for barley to determine the activation energy for all the temperatures recorded on the study site.

The modeling process generates 35 distinct LRCs, each corresponding to temperatures ranging from 0 °C to 35 °C. For each curve, a KP is identified, representing the optimal PAR at that specific temperature. These discrete values are then interpolated to derive a continuous profile of optimal PAR across all hourly time steps during the crop growth period. At each time step, this temperature-dependent optimal PAR is compared to simulated ground-level irradiance values across all tilt angles. The tilt angle that produces irradiance closest to the optimal PAR is selected. This method provides a streamlined and adaptive strategy for dynamic tilt angle selection, using ambient

temperature as the sole input for optimal PAR estimation. Fig. 4 shows the schematic for the KPT model.



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Fig. 4. Schematic diagram showing the flow of the KPT model.

2.4. Biomass accumulation

Biomass accumulation during plant growth can be estimated by integrating photosynthetic carbon assimilation over time. This involves measuring the net photosynthetic CO₂ uptake per unit area and per second, then converting it into grams using the molar mass of carbon (12 g mol⁻¹). To determine the actual net carbon gain, it is essential to account for plant respiration. Both maintenance respiration and growth respiration are subtracted from the total assimilated carbon. According to Amthor [63], the combined respiration for cereals typically ranges between 30 % and 60 % of the gross photosynthesis rate. After calculating the net carbon gain, it is then converted into dry biomass using the carbon content of the plant. For barley, the carbon content in biomass is generally around 45 % [64,65]. This method provides a biologically grounded estimate of biomass production and is consistent with empirical models and established

principles of C3 cereals photosynthesis, offering a reliable approach to linking photosynthetic activity with biomass accumulation. (9) where BM is biomass accumulation (ton ha^{-1}), P is the photosynthetic rate ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$), M is molar mass of carbon (g mol^{-1}), RF is the respiration fraction including maintenance and growth respiration (60 %), and CF is the carbon fraction of the total biomass (45 %). These values represent average parameters reported for barley under European conditions [64], although RF and CF may vary with cultivar and environment, reasonable deviations around these values are not expected to alter the comparative ranking of the tracking strategies.

The total biomass for a sample area is initially calculated in kilograms per square meter (kg m^{-2}) and then scaled up to one hectare by multiplying by 10^4 , as shown in Eq. (9). The molar mass of carbon (12 g mol^{-1}) is used to determine the amount of carbon associated with each measured photosynthetic rate and is obtained after conversion from $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ by the factor [66]. To account for plant respiration, a maximum respiration factor of 60 % of the total assimilated carbon is applied [63]. The fraction of land area rendered unusable due to the piles supporting the PV structure is denoted as LL in Eq. (9). Specifically, a strip of 0.5 m on both the east and west sides of each pile cannot be cultivated, resulting in approximately one-tenth of the land being unsuitable for barley cultivation.

2.5. Applicable tilt angle

Rapid fluctuations in PAR can lead to sudden changes in the optimal tilt angle of solar modules. Consequently, the system may rapidly transition from a sun-tracking mode, aimed at maximizing ground irradiance, to a configuration that intentionally reduces ground irradiance. These abrupt shifts often require larger angular movements between time steps, prompting the tracker motor to operate more frequently and for longer durations. This increased activity may result in higher mechanical stress and wear on the tracking components, as well as greater energy consumption. To address this issue, Bruno et al. [32] proposed the use of a Gaussian filter to smooth the tilt angle adjustments at hourly intervals. Following this approach, the current model, deploys the `Gaussian_filter1d` function from the SciPy library to achieve a similar smoothing effect with standard deviation of Gaussian Kernel 1 [67]. This method works by replacing each data point with a weighted average of its neighbouring points, where closer neighbours have more influence than distant ones as seen in Eq. (10). The 'spread' of this influence is controlled by the standard deviation of the Gaussian curve, in this case, a value of 1, which results in moderate smoothing without losing too much detail. The outcomes, including biomass accumulation and overall energy yield, can then be compared to those obtained without applying the smoothing technique to evaluate the trade-offs.

In general the Gaussian Filter is implemented mathematically as described in Johannes [68]:(10) In Eq. (10), θ_t is the corresponding Gaussian-smoothed tilt angle applied to the tracker, θ_{t-k} denotes the unsmoothed tracker tilt angle at time step k as determined by the selected tracking strategy (TT, AT, ST, DLIT, or KPT), and θ_t is the corresponding Gaussian-smoothed tilt angle applied to the tracker. The index k represents the discrete time-step offset relative to t , while w defines the half-width of the smoothing window such that $2w+1$ neighboring time steps are included. The parameter σ is the standard deviation of the Gaussian kernel, controlling the degree of smoothing (here $\sigma = 1$), and the weighting term assigns higher weights to time steps closer to t and lower weights to more distant neighbors.

3. Results

This section presents the results of the proposed tracking strategies, focusing on the optimal PAR values derived from LRCs across a range of temperatures. These optimal PAR values are used as decision variables in the KPT approach. The analysis includes both conventional tracking strategies and a novel framework that incorporates DLI and KP criteria to guide tilt adjustments. To evaluate the effectiveness of each method, tilt angle trajectories over the course of a representative day are analyzed alongside the corresponding biomass accumulation. The final part of this section examines how smoothing the tracking trajectories influences both energy yield and crop productivity. This comparative assessment enables a deeper understanding of how well each strategy balances energy conversion with crop light requirements, providing insights into their practical applicability in APV systems.

3.1. Optimal PAR

Fig. 5 illustrates the identified KPs on the LRC across different temperatures, determined using the KP detection method. This method pinpoints the optimal PAR value, where the rate of photosynthesis begins to level off, based on the physiological light response characteristics specific to barley. These KPs represent the transition zone beyond which further increases in light intensity result in diminishing gains in photosynthetic rate.

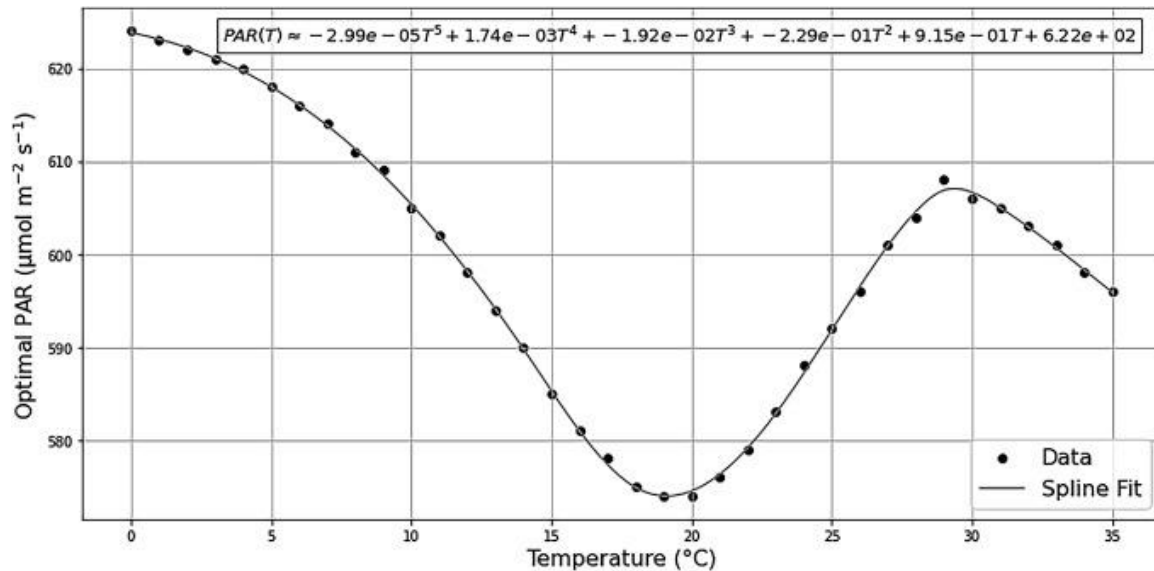


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Fig. 5. Photosynthetic rate for PAR (0 to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$) at different temperatures ranging from 0 to 35°C, with the KP where the optimal PAR occurs.

This method establishes a clear and continuous relationship between optimal PAR and temperature, enabling the estimation of temperature-specific PAR thresholds that inform strategic decisions on whether to prioritize crop productivity or energy conversion. Fig. 6 demonstrates an extended curve that fits the photosynthetic response data across all temperature ranges. By applying this fitted curve, the optimal PAR can be accurately predicted for any given temperature, serving as a key input for the proposed tracking strategy to dynamically balance agricultural and energy objectives.



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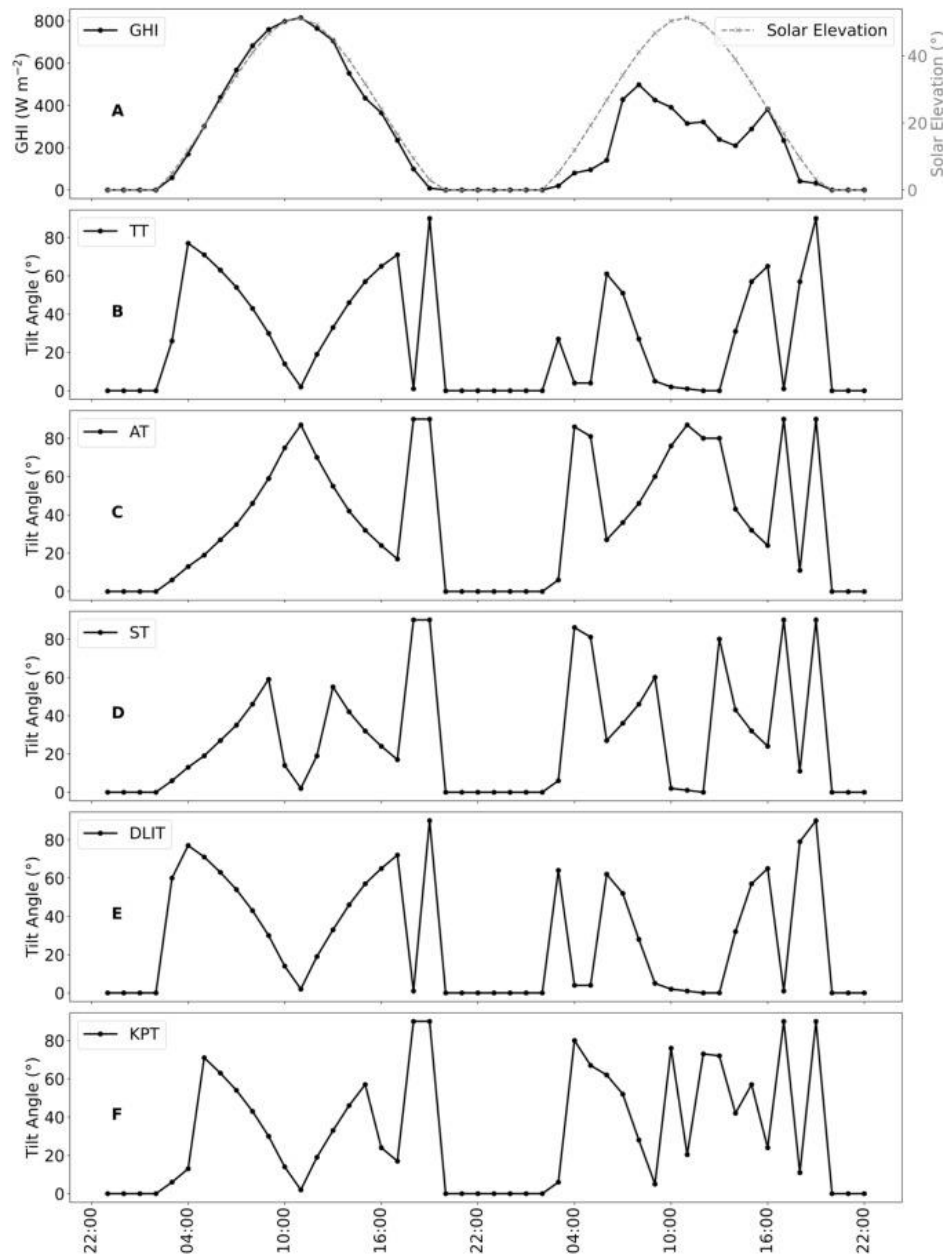
Fig. 6. Optimal PAR for any given temperature.

The black dots represent observed optimal PAR values obtained using the KP detection method across a range of temperatures. The gray curve represents the spline fit, which models this relationship with a high degree of accuracy. The polynomial equation shown describes the fitted curve, allowing for the estimation of optimal PAR at any given temperature. This fitted model serves as a critical input for adaptive tracking strategies that balance crop productivity and energy conversion in a dynamic APV system.

In this study, adequate radiation is defined physiologically, either as instantaneous PAR values near the KP of the LRC or as cumulative daily irradiance sufficient to meet the crop-specific DLI requirement.

3.2. Tilt angle dynamics for different tracking strategies

Fig. 7 presents the variation in tilt angles for different solar tracking strategies over a two-day period, from May 24 to May 25, 2019, alongside the corresponding solar elevation angle. The solar elevation is displayed on the secondary y-axis (right), while subplot (A) shows the GHI on the primary y-axis (left). Subplots (B) through (F) depict the tilt angle profiles for five distinct tracking methods: TT, AT, ST, DLIT, and KPT. Each subplot illustrates how the respective strategy responds to changing irradiance conditions by adjusting the tilt angle dynamically throughout the day, providing insights into the operational behavior and responsiveness of each approach under different sky conditions.



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Fig. 7. Hourly tilt angle variations under different tracking strategies for a clear-sky day (May 24, 2019) and a cloudy day (May 25, 2019). All subplots illustrate: Solar elevation vs. GHI (A), TT (B), AT (C), ST (D), DLIT (Daily) (E), and KPT (F).

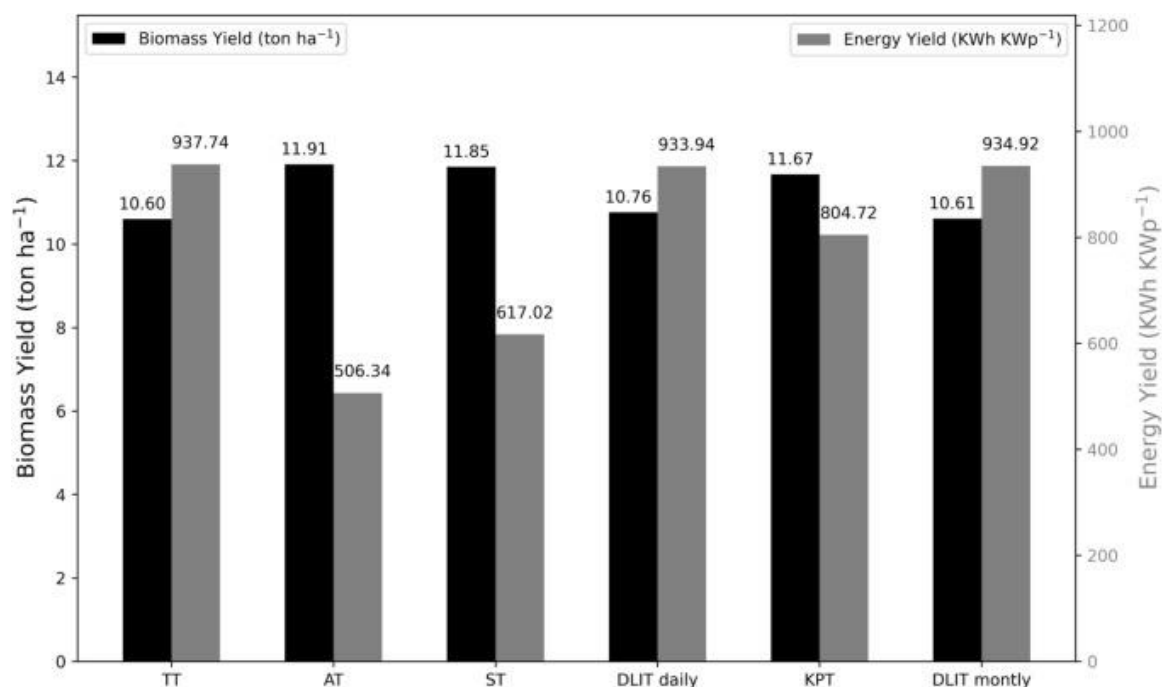
The character of the available GHI on a given day significantly influences the smoothness of tilt angle transitions in solar tracking systems. On clear-sky days, the irradiance follows a predictable and continuous pattern, resulting in smoother and more

gradual changes in the tilt angle throughout the day. In contrast, on cloudy or intermittently overcast days, GHI fluctuates due to frequent changes in cloud cover. This irregularity causes the tracking system to make frequent and abrupt tilt adjustments in response to the varying solar input, leading to greater instability in tilt angle trajectories.

As illustrated in [Fig. 7](#), May 24, 2019 represents a clear-sky day, as shown in subplot A, where GHI follow a smooth profile. In comparison, May 25, 2019 is characterized by cloudy conditions, reflected in the erratic GHI pattern. Consequently, all tracking strategies demonstrated more frequent and pronounced variations in tilt angle on May 25, 2019, while the tilt profiles remained stable and consistent on May 24, 2019. This contrast highlights the sensitivity of tracking systems to daily irradiance conditions and their implications for operational stability.

3.3. Biomass yield vs energy yield

The comparative performance of five tracking strategies in an APV system, with a focus on biomass yield and energy yield, is presented in [Fig. 8](#). The strategies tested include TT, AT, ST, DLIT, and KPT. TT achieved the highest energy yield at 937.74 kWh kWp⁻¹, but the lowest biomass yield at 10.60 ton ha⁻¹, reflecting its focus on maximizing energy conversion at the cost of crop growth. In contrast, AT maximized biomass yield at 11.91 ton ha⁻¹, but had the lowest energy yield at 506.34 kWh kWp⁻¹, highlighting the trade-off between energy and agricultural productivity. ST provided a balanced outcome with 11.85 ton ha⁻¹ of biomass and 617.02 kWh kWp⁻¹ of energy yield, favoring crop growth. KPT, which adjusts module tilt based on optimal PAR, achieved 11.67 ton ha⁻¹ of biomass and 804.72 kWh kWp⁻¹ of energy, offering a better balance between energy conversion and crop growth. DLIT, which adapts based on daily crop light requirements, yielded 10.76 ton ha⁻¹ of biomass and 933.94 kWh kWp⁻¹ of energy. When used a monthly average light requirement (as in Bruno et al. [32]) DLIT achieved 10.61 ton ha⁻¹ and 934.92 kWh kWp⁻¹, with minimal variation from the daily-based setup, as shown in [Fig. 8](#).



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Fig. 8. Comparison of energy yield and biomass yield across five tracking strategies implemented in the APV system: TT, AT, ST, DLIT, and KPT. The DLIT strategy was tested in two configurations: one using daily DLI thresholds, and the other using monthly average DLI thresholds for each month from April to August.

The results in Fig. 8 highlight the trade-offs between biomass and energy yields across the evaluated strategies. Following the overview in Fig. 8 and Table 3a, Table 3b, it becomes evident that the AT strategy not only achieved the highest biomass yield (11.91 ton ha⁻¹) but also served as the primary reference point for evaluating biomass performance across the strategies. Both ST and KPT performed closely to AT, with only minor reductions in biomass yield (0.51 % and 2.02 %, respectively), which supports their potential as more balanced alternatives without significantly compromising crop growth. On the other hand, the TT and DLIT strategies, showed lower biomass yields, aligning with their performance trends described earlier. In terms of energy yield, the detailed comparison confirms TT as the top-performing strategy, with the DLIT approaches closely following and differing by less than 0.5 %, consistent with their strong energy focus shown in Fig. 8. In contrast, AT, ST, and KPT delivered substantially lower energy yields, reinforcing the observed trade-off: strategies favouring biomass production inherently limit energy output, while those maximizing energy tend to reduce crop productivity.

Table 3a. Biomass yield comparisons benchmarking the AT yield (ton ha⁻¹).

Strategy	Yield	Absolute difference	% difference
TT	10.60	-1.31	-11.00
AT	11.91	0.00	0.00
ST	11.85	-0.06	-0.51
DLIT-Daily	10.76	-1.15	-9.60
KPT	11.67	-0.24	-2.02
DLIT-Monthly	10.61	-1.30	-10.91

Table 3b. Energy yield comparisons benchmarking the TT yield (kWh kWp⁻¹).

Strategy	Yield	Absolute difference	% difference
TT	937.74	0.00	0.00
AT	506.34	-431.40	-46.00
ST	617.02	-320.72	-34.20
DLIT-Daily	933.94	-3.80	-0.40
KPT	804.72	-133.02	-14.18
DLIT-Monthly	934.92	-2.82	-0.30

These observed trade-offs are consistent with previously reported relationships between PV system design and under-module light availability [69]. Systems that prioritize maximizing solar energy capture by the PV modules often result in reduced light transmission to the ground, which can limit crop biomass production. In contrast, configurations that permit greater light penetration to the crop canopy tend to support higher biomass yields, though this typically comes at the expense of PV electricity generation [8,70].

The result further indicates that there is no significant difference between using a daily based DLI threshold and the monthly average DLI threshold, suggesting that using monthly averages, as done in previous studies focusing solely on monthly PAR targets [32], is a valid approach without substantial loss of accuracy. However, it should be noted that the results obtained from the two novel approaches are crop-specific to barley. Energy and biomass outcomes may vary for other crop types, as the aim of this study is to compare tracking strategies rather than differences between crops.

Therefore, Fig. 8 shows that DLIT achieves higher energy yield than KPT but slightly lower biomass, because once the crop-specific DLI target is satisfied, the algorithm

shifts toward energy-oriented (TT-like) operation. In contrast, KPT continuously follows the PAR level corresponding to the knee of the LRC, which maintains higher photosynthetic efficiency and results in greater biomass at the expense of some electricity production.

3.4. Overall comparison before and after smoothing

The impact of tilt angle smoothing on energy and biomass yield was evaluated by comparing the performance of all tracking strategies before and after applying the smoothing process. Table 4a, Table 4b presents a summary of the biomass and energy yield values, along with the relative percentage differences. Across all strategies, biomass yield decreased significantly following smoothing, with reductions ranging from approximately 22 to 24 %. This consistent decline highlights the sensitivity of crop growth to the ability of the PV tracking system to dynamically adjust tilt angles to maximize light availability. Rapid tilt adjustments in the unsmoothed data enable the system to better track optimal solar angles and thereby allow more light to reach the crop. When these rapid changes are dampened through smoothing, the system's responsiveness is reduced, leading to lower ground-level irradiance and ultimately diminished biomass accumulation.

Table 4a. Biomass yield comparison for the smoothed and unsmoothed tracking tilt angle for all tested tracking strategies (ton ha⁻¹).

Strategy	Smoothed	Unsmoothed	Absolute difference	% difference
TT	8.22	10.60	2.38	-22.45
AT	9.07	11.91	2.84	-23.84
ST	9.03	11.85	2.82	-23.79
DLIT	8.39	10.76	2.37	-22.02
KPT	8.98	11.67	2.69	-23.05

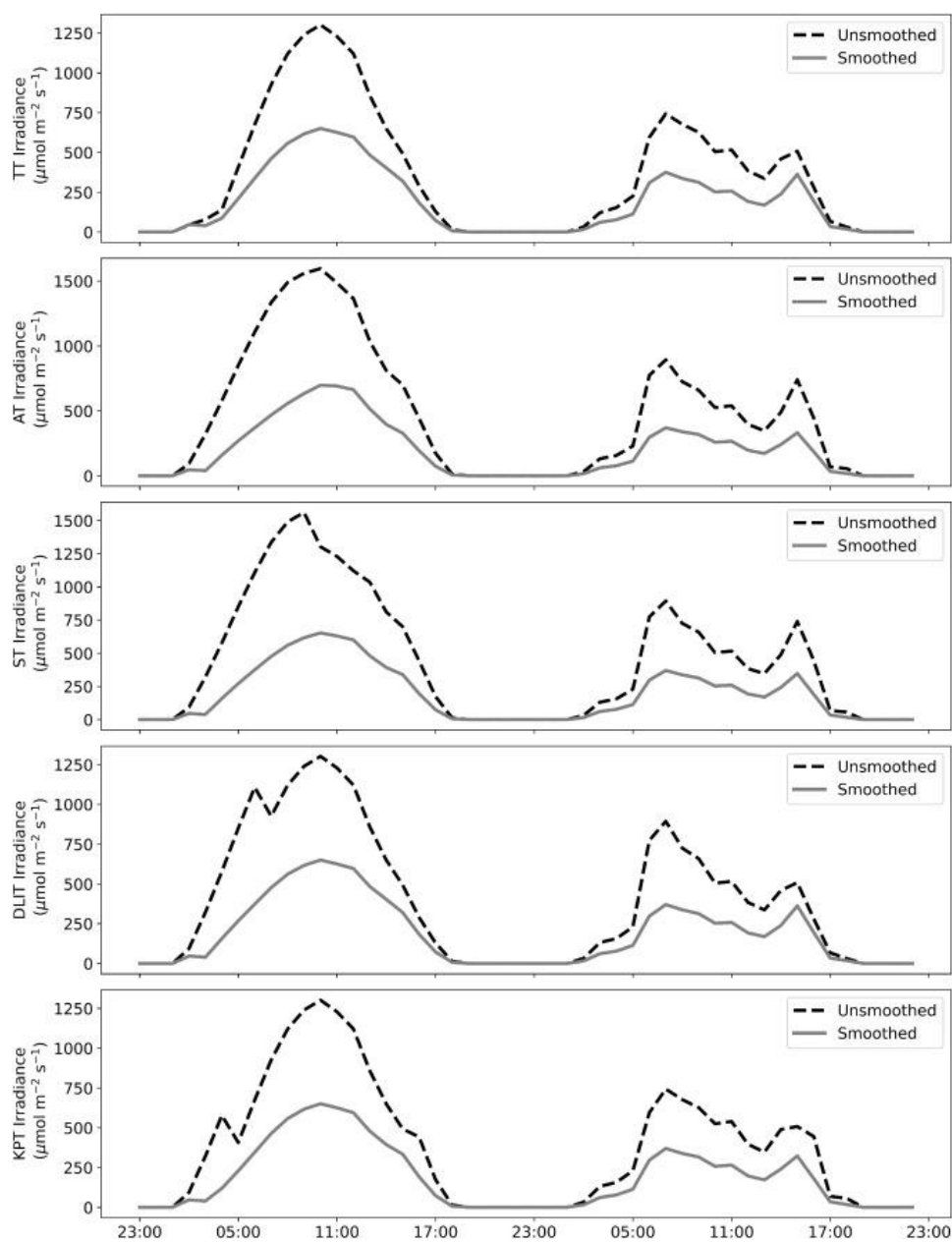
Table 4b. Energy yield comparison for the smoothed and unsmoothed tracking tilt angle for all tested tracking strategies (kWh kWp⁻¹).

Strategy	Smoothed	Unsmoothed	Absolute difference	% difference
TT	869.80	937.74	67.94	-8.50
AT	616.94	506.34	110.6	+19.6
ST	702.72	617.02	85.7	+12.9
DLIT	869.87	933.94	64.07	-7.52

KPT	800.18	804.72	4.54	0.56
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In contrast, the energy yield response to smoothing varies by strategy. The TT, DLIT, and KPT approaches showed moderate decreases in energy yield, with TT and DLIT experiencing reductions of approximately 7.5 to 8.5 %, while KPT remained unchanged. This suggests that smoothing reduces the system’s ability to precisely align the module for optimal energy conversion in some configurations, but not all.

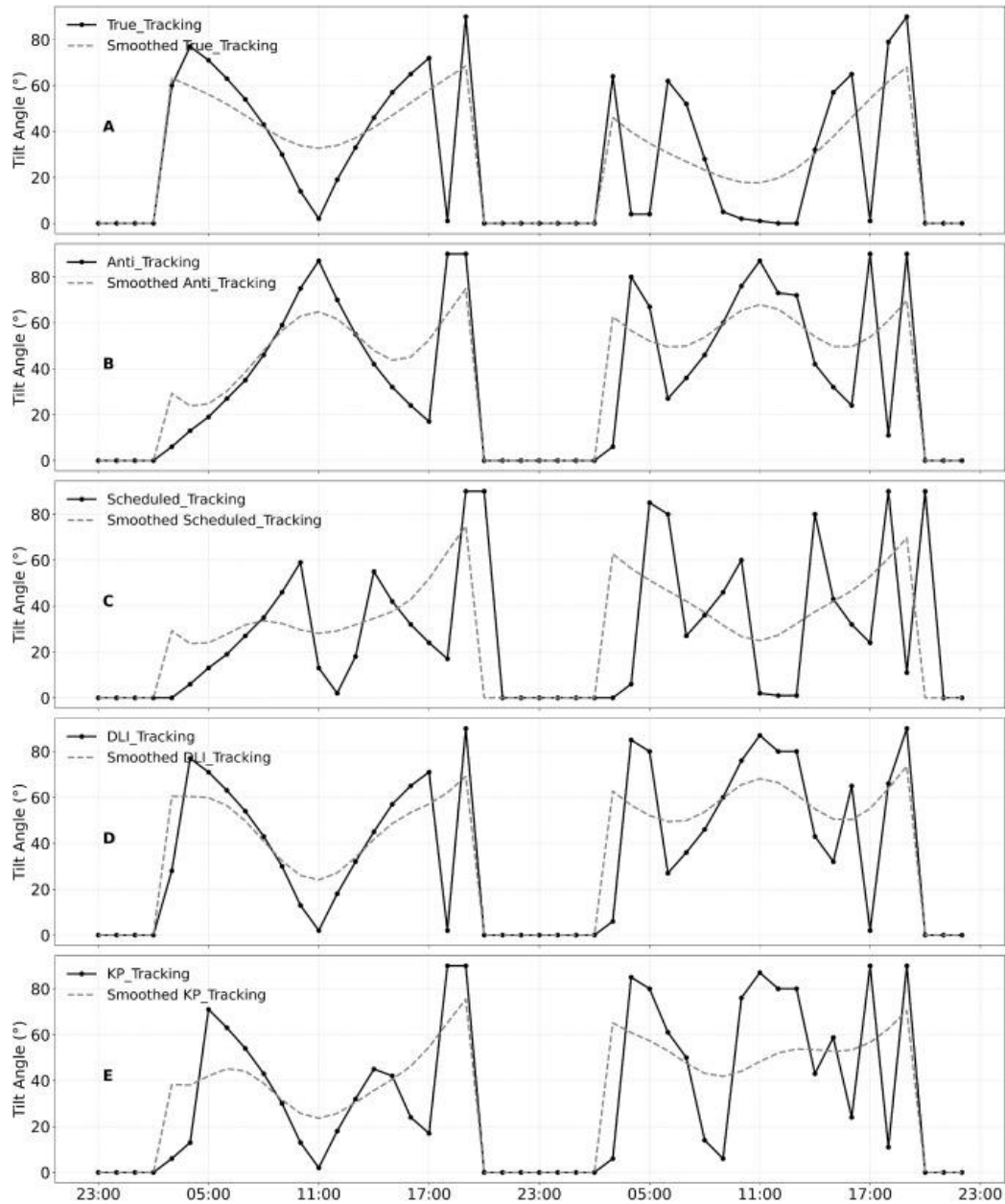
Interestingly, the AT and ST strategies exhibited moderate increases in energy yield after smoothing, by about 12.9 and 19.6 % respectively. This counterintuitive outcome may result from the smoothing process eliminating overly aggressive or suboptimal tilt fluctuations present in the unsmoothed data, thus enabling a more stable and efficient energy conversion profile. [Fig. 9](#) visually demonstrates the effect of smoothing on ground-level irradiance patterns for each tracking strategy. The unsmoothed data shows sharper fluctuations in irradiance corresponding to rapid tilt angle changes as depicted in [Fig. 10](#). After smoothing, these fluctuations are dampened, producing a more gradual irradiance profile but limiting the system’s ability to exploit transient periods of optimal light availability. This mechanistic understanding explains the observed reductions in biomass yield and the mixed impacts on energy yield.



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Fig. 9. Ground irradiance before and after smoothing for all tracking strategies for the 24 and 25 May 2019.



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Fig. 10. Smoothed vs unsmoothed tracking tilt for all strategies for 24 and 25 May 2019.

The effect of smoothing on energy yield differs among strategies because each strategy exhibits distinct tilt-angle fluctuation patterns prior to filtering. Strategies with frequent directional changes, such as DLIT, experience a larger reduction in energy after smoothing, whereas TT and AT, which already follow relatively stable trajectories, show minimal sensitivity.

4. Discussion

4.1. Energy-biomass trade-offs

The tilt angle of the PV system plays a crucial role in balancing energy conversion and biomass yield in APV systems. Generally, tilt angles that maximize ground-level irradiance support biomass accumulation, while those that minimize ground irradiance tend to favor energy conversion by increasing the irradiance captured by the PV modules. However, this relationship is not always straightforward. As demonstrated in this study, aligning the module surface perpendicular to the solar beam, as in the traditional TT method, does not consistently result in the highest energy yield under all weather conditions. For example, on cloudy days, a slightly offset tilt angle may perform better due to the diffuse nature of solar irradiation, deviating from the benefits typically associated with direct sun tracking [71]. This dynamic is evident in Fig. 7 where May 24, 2019, a clear-sky day, shows smooth and predictable tilt adjustments aligned with the solar position or its inverse, as observed in the AT strategy. On May 25, 2019, a cloudy day, more frequent and irregular tilt fluctuations occurred across most tracking strategies. These variations reflect the system's adaptive response to rapidly changing irradiance conditions, suggesting that the optimal tilt angle does not always correspond to the sun's direct position.

When comparing energy yield across the evaluated tracking strategies, the AT method resulted in a 45.74 % reduction relative to the TT baseline, which is consistent with the findings of Alam and Butt [33], who reported 40–50 % energy conversion reductions for AT compared to TT. Similarly, Grubbs et al. [25] evaluated a seasonally controlled tracking strategy that alternated between TT and AT according to crop growth stages and reported a 29.67 % reduction in energy yield relative to TT. The ST approach implemented in this study, which is functionally comparable to such stage- or time-controlled TT/AT systems, resulted in a 34.04 % energy conversion reduction relative to TT. In this case, the timing of TT and AT within the ST strategy follows the diurnal scheduling proposed by Valle et al. [26], where TT is applied during midday hours (11:00–15:00) and AT is used outside this window to mitigate excessive crop shading.

In contrast, the novel crop-responsive tracking strategies proposed in this study, DLIT and KPT, demonstrate substantially improved energy performance relative to AT and ST. The DLIT strategy achieved near-TT energy yield, with a reduction of less than 0.5 %, while the KPT strategy resulted in a more moderate 14 % reduction compared to TT. Both approaches significantly outperform conventional crop-prioritizing strategies in terms of energy conversion. These energy gains, however, involve different trade-offs in biomass accumulation. DLIT, despite its strong energy performance, led to a 9.6 %

reduction in biomass, reflecting its shift toward energy-oriented operation once the crop-specific $DLI(x)_{\text{target}}$ is satisfied. Conversely, KPT prioritized biomass production by maintaining irradiance near the KP of the LRC, accepting a moderate energy penalty to enhance crop growth. Notably, KPT also outperformed DLIT in terms of biomass yield, highlighting its potential as a more balanced tracking strategy for APV systems.

Other studies have sought to address the energy–crop trade-off through alternative control approaches. For example, Bruno et al. [32] developed an optimization-based tracking method that dynamically adjusts tilt angles to maintain ground-level irradiance close to predefined targets. While their approach did not provide a direct comparison with standard tracking strategies such as TT or AT, a meaningful parallel can be drawn between their use of monthly irradiation targets and the DLIT framework proposed in this study. The close agreement observed here between daily and monthly DLI-based implementations further supports the practical relevance of such target-based approaches.

From an agronomic perspective, the simulated biomass yields obtained in this study are consistent with reported benchmarks for barley. Rogers et al. [72] reported harvest index values ranging from 0.46 to 0.52, while Morel et al. [73] documented typical barley grain yields in Sweden between 4.50 and 5.80 ton ha^{-1} . When the simulated total biomass values in this study (10.60–11.91 ton ha^{-1}) are multiplied by the harvest index reported by Rogers et al. [72], the resulting grain yield estimates fall within the expected range (4.87 - 5.47 ton ha^{-1}) reported for Swedish conditions. This provides indirect validation of the modeling framework and supports the agronomic plausibility of the simulated energy–biomass trade-offs.

4.2. Practical implications and industrial applications

Although the quantitative results reported in this study are specific to barley and to the APV configuration at Kärrbo Prästgård, the two novel strategies DLIT and KPT are formulated as generic control frameworks rather than crop dependent rules. DLIT can be applied to other crops by replacing the LAI-based DLI profile with crop-specific light requirements, while KPT only requires the corresponding light-response curve to determine the temperature-dependent optimal PAR. For crops with lower light demand, such as leafy vegetables, the algorithms would tend to favor TT-like behavior for longer periods, whereas for shade intolerant crops would approach AT-like operation. Therefore, the methodological approach is adaptable across common APV crops, although the optimal thresholds and performance gains must be recalibrated for each crop and climate.

Smoothing the tracking tilt presents an opportunity to improve the mechanical reliability of APV systems by minimizing abrupt actuator movements driven by transient weather fluctuations, such as passing clouds. More gradual tilt adjustments can help reduce wear on tracker components, potentially lowering maintenance frequency and extending system lifetime [20]. However, this operation advantage comes at the cost of reduced ground-level irradiance, as shown in Fig. 9, which can negatively impact biomass accumulation across all tracking strategies as the results indicate in Table 4a. This highlights an important trade-off for APV system designers: maximizing mechanical longevity may conflict the goal of maximizing crop productivity.

The crop-responsive tracking algorithms developed in this study, particularly DLIT and KPT, offer significant potential for commercial APV installations and large-scale solar farms. By dynamically adjusting module tilt to prioritize either energy yield or crop light requirements, these strategies enable more efficient land use and dual optimization of electricity production and agricultural output. Importantly, their inherent flexibility allows for easy adaptation to diverse national APV regulations, such as those in France, Germany, and Italy, ensuring regulatory compliance while maintaining high system performance [[74], [75], [76]].

Implementing these strategies in industrial settings will require advanced system designs that incorporate real-time sensor networks. Additionally, control algorithms must be capable of processing real-time data streams to make responsive tilt adjustments under varying weather conditions. Future industrial applications should also consider integrating forecasting tools and predictive controls to further optimize tracking behavior, ensuring that tilt adjustments are not solely based on immediate sensor readings but also account for expected weather changes. Such integration would enable APV system to proactively manage the trade-offs between energy yield, crop performance, and mechanical stress.

From a sustainability perspective, the proposed DLIT and KPT strategies directly support the SDG objectives discussed in the Introduction by enabling simultaneous optimization of crop productivity and renewable electricity generation within the same land footprint. By reducing the trade-off between food and energy conversion at the system level, these strategies enhance land-use efficiency and strengthen the practical contribution of APV systems to food security, clean energy deployment, and climate mitigation.

Overall, the results demonstrate the practical viability of the proposed DLIT and KPT strategies for APV systems. Their performance is evaluated under realistic site conditions, mechanical constraints, and crop-physiological limits, and benchmarked against commonly applied tracking strategies. The explicit consideration of controller

stability, smoothing effects, and implementation requirements further supports the applicability of the proposed approaches in real-world APV installations.

4.3. Limitations of the study and suggestions for future research

One important limitation of this study is that, although the effect of applying a Gaussian filter to smooth rapid tilt-angle adjustments under unstable weather conditions on biomass accumulation and energy yield are quantified, the resulting reductions in actuator energy consumption or the potential extension of tracker component lifetimes are not evaluated. While dampening frequent movements may lead to meaningful operational savings and reduced mechanical wear, these benefits remain unquantified. Future assessments should therefore combine yield analyses with detailed evaluations of actuator power demand, maintenance requirements, and component fatigue life.

The study is entirely simulation-based, as field data from the APV site will become available later than anticipated and will be used for validation in forthcoming work. The current results rely on modeled LAI development, photosynthetic parameters, and derived DLI targets based on existing literature for crops grown under open-field conditions rather than from direct measurements under APV. Local factors such as soil conditions, nutrient status, water availability, agronomic management and cultivar differences are not explicitly represented and may influence actual crop responses under APV shading. Field validation with in-situ measurements of canopy development, photosynthesis, and yield is therefore essential to confirm the practical performance of the proposed strategies.

Another limitation is that the tracking algorithms assume perfect knowledge of weather conditions because historical hourly data were used to simulate system operation. In real applications, forecasting errors and sensor uncertainties would likely affect tilt decisions and system performance. The integration of predictive control methods and uncertainty-aware tracking strategies is required to ensure robust operation under real-time conditions.

Biomass estimation is based on fixed respiration of 60 % and carbon fraction of 45 % as these parameters are applied uniformly across all tracking strategies, variations around these values would mainly scale absolute biomass estimates without altering the comparative ranking among strategies. Nevertheless, future work should incorporate stage-dependent and cultivar-specific coefficients derived from field measurements to improve absolute yield prediction of biomass.

The present analysis is strictly focused on energy and biomass indicators and does not explicitly represent the techno-economic implications, including investment costs,

operational expenses, and revenue trade-offs between electricity and crop production as they were outside the scope of this study.

Finally, the quantitative findings are specific to barley and to the studied APV configuration in Sweden. Although the DLIT and KPT frameworks are conceptually transferable to other crops, climates, and system geometries, optimal thresholds and performance gains may differ. Multi-crop field trials across diverse locations and PV layouts are therefore required before the strategies can be generalized for large-scale commercial deployment.

5. Conclusion

This study evaluated two novel crop-responsive single-axis tracking strategies DLIT and KPT against conventional APV benchmarks, namely TT, AT, and ST, under Swedish conditions. The central objective was to assess whether explicitly incorporating crop light demand into tracker control can improve the balance between PV energy conversion and barley biomass yield.

Benchmarking confirmed the characteristic extremes of conventional strategies, TT maximized electricity supply but resulted in the lowest biomass, while AT achieved the highest biomass at the expense of nearly half of the energy yield relative to TT. ST provided an intermediate but static compromise. In contrast, the proposed crop-aware strategies DLIT and KPT demonstrated a substantially improved energy–biomass balance. DLIT maintained energy conversion within approximately 1 % of TT while reducing biomass by about 10 %, indicating that daily crop light constraints can preserve energy output with limited agronomic penalties. KPT further improved this balance by limiting biomass losses to around 2 % relative to AT while retaining more than 85 % of the energy yield achieved by TT.

The analysis also revealed that smoothing tilt-angle trajectories reduced biomass yields by more than 20 % across all strategies, underscoring the importance of dynamic and responsive tracker operation for crop performance. Overall, the key advantage of the proposed DLIT and KPT frameworks lies in their physiological grounding and operational flexibility. By directly linking tracker control to cumulative and instantaneous crop light requirements, these strategies move beyond fixed schedules and empirical thresholds, offering a robust pathway toward improved dual-use land efficiency in APV systems.

CRediT authorship contribution statement

Sultan Tekie: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sebastian Zainali:** Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Silvia Ma Lu:** Writing – review & editing, Methodology. **Mohammed Guezgouz:** Writing – review & editing, Supervision. **Tekai Eddine Khalil Zidane:** Writing – review & editing, Visualization, Supervision. **Arash Khosravi:** Writing – review & editing. **K. Max Zhang:** Writing – review & editing. **Pietro Elia Campana:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

The company European Energy AB is financing half of Sebastian Zainali's Ph.D. salary.

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